INMT5526: Business Intelligence
Group Assignment – Semester Two, 2025

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Exploring Barriers and Enablers of Digital Financial Inclusion through Business Intelligence

Executive Summary :

To find out the reasons behind the fact that after 10 years of global financial access development, there are still horrendous discrepancies in respect of the income level of the individual, the region, and the sex, the project assesses Digital Financial Inclusion (DFI) with the help of Global Findex Database (2014-24). While over 60% of adults worldwide report owning an account, only about half actively use digital payment. It is an activation gap that has existed long enough and restricts the promise of financial inclusion through this disparity between access and active use. Income and region are still the main factors determining the situation, but the structural drivers like the wage disbursement into accounts and the presence of e-payment systems can be pointed out as equally significant by our analysis.

To investigate this problem, the dataset was first cleaned to remove nulls and inconsistencies, then dimension tables (country, region, income, year, gender) were created for better modeling. Many composite measures were introduced, among them being DFI, Account vs Usage gaps, Wage-to-Account, Digital vs Cash preference, and peer benchmarking measures (Rank and Lift vs Income Median). Power BI dashboards were visualized with KPI cards, regional/world maps, income and region bar charts, scatterplots for usage drivers, and ranking tables to compare countries against their peers through the visualization of these measures. Some key conclusions drawn from the analysis include:

Digital adoption is led by the high-income countries , while low-income countries show a significant gap in this regard, some regions show limited progress beyond account ownership, reflecting structural barriers and wage payments into accounts are a key lever strongly correlated with digital usage. The benchmarking demonstrates that certain countries either outperform or underperform their income peers, hence making it clear that policy and infrastructure decisions have an impact beyond income level.

To conclude, the project shows that the concept of access does not necessarily imply inclusion. The way to realize sustainable digital financial inclusion is to ensure that countries

concentrate to turn account ownership into regular digital use, grow digital payment systems, and create long-term financial resilience. The Power BI dashboards developed during this study can be used as a strong policy-monitoring instrument enabling governments, development institutions and researchers to observe the progress, compare their performance with peer groups, and recognize the best practices to be replicated and inclusively to promote financial development across the globe.

Introduction :

The current digitalization of world financial systems has transformed the way people and establishments handle, transmit and maintain money. Digital Financial Inclusion (DFI) refers to the practice of making sure that everyone, particularly the already stigmatized segment of the population, has access to and the capacity to utilize financial services provided via digital means [1]. It goes beyond having a bank account to make payments, saving, borrow and insuring in a safe, low-cost and sustainable way using digital technology. DFI has become a vital part of an inclusive growth, which also makes possible the economic participation, women empowerment and financial resilience in the developing nations [2], [9].

Although there has been significant global advancement on financial inclusion, there is still uneven digital financial inclusion amongst income groups, regions, and gender. In the database on Global Finance Index provided by World Bank (2014-2024), 60 per cent of adults all over the world are reported to have a financial account, out of which only half is actively using the account to make digital payments [7]. The continual activation gap points to the fact that one thing is access the other that the usage can be meaningful. The rates of digital payment adoption are more than 80 in high-income countries, and below 35 in low-income countries. Likewise, other parts of the world like Sub-Saharan Africa and South Asia, even though they have been increasing the number of account owners are still lagging in the field of digital payments adoption [3], [7].

Accessibility and utilization disparity is a conglomeration of several structural and socioeconomic elements. Based on the Findex data, which we analyze, we observe that the payment in the accounts of wages and the availability of digital payment infrastructure are the best predictors of digital engagement [5]. Where digital payment of wages and benefits is done and made by payers through employers and governments, the opportunities of further continued usage are hopelessly high. Quite on the contrary, the account dormancy is high where access to the internet is low and where people lack confidence in the online processes. In addition, the financial resiliency behaviors including saving and borrowing remain trapped at approximately 50 percent suggesting that online access to finance may not necessarily result into long-term financial stability [6].

The problem by that point is no longer in increasing access, but in getting the ownership in action and stable and ensuring that the inclusion is empowerment. When the median incomes are compared with the median incomes of the income groups, the countries are performing either better or worse than their counterparts, which implies that policy choices and institutional set up, as well as infrastructure, are influencing factors irrespective of the level of income [1], [3]. In the absence of specific work to raise digital literacy and payment rails and build confidence of the financial ecosystem, vast layers of the global population in the low and middle-income economies, are at risk of remaining not beneficiaries of the digital transformation.

Significance of Digital Financial Inclusion

DFI has an important role in achieving sustainable development goals related to alleviating poverty, ensuring gender equality and economic resilience. It reduces the transaction costs, increases transparency and broadens access to credit by marginalized people. According to IFAD (International Fund for Agricultural Development), digital remittance and mobile money have enabled faster, cheaper, and safer remittance of millions of rural dwellers [2]. As shown by the case of Vietnam, say, mobile money, digital instruments can directly increase engagement of the unbanked and underbanked people [4]. Nevertheless, multiple studies by World Bank Executive Independent Evaluation Group (IEG) caution the world that most inclusion initiatives are not that effective either in terms of account creation or sustained engagement or capacity building [3]. Real DFI must not only be technological solutions, but behavioural and institutional changes that result in active use and longer term savings behaviour [5].

Project Objectives

This project will explore and visualize inequality within the financial inclusion in the digital world compared to countries using the Global Findex Database (2014-2024). The main objectives are to:

1. Segment the digital financial inclusion by socio-economic groups, region, and gender.
2. Gap analysis between access and usage based on data modeling and visualization using Power BI.
3. Analyse the impact of structural factors on adoption e.g. wages payments, digital infrastructure, and policy environments.
4. Compare countries in the income group with one another to determine which countries are over performing and which are underperforming.
5. Provide validated insights to the policymakers for sustainable and equitable digital inclusion.

This project delivers a data-driven model to monitor DFI's performance with the help of visualization of dashboards. The results underline the necessity for countries and financial institutions to consider not just improving account access, but also launching active engagement in digital practice in addition to driving wage digitization and financial empowerment over the long-term for people.

Methodology :

The GlobalFindexDatabase2025 was loaded into Power BI, and the data transformation began. The dataset contains 437 columns and 8566 observations. For the data cleaning, initially, a null threshold of 0.5 was set to drop all columns that surpassed this threshold. After this step, the 437 columns dropped to just 28 columns. Then the data types of the float values were changed from text to decimal type. After which, the columns were renamed by referencing the glossary provided. A table with all the years, namely, dimyear was created with all the years of the dataset and joined with the GlobalFindexDatabase2025 dataset in a 1:* relationship. Another table, namely dimcountry was created with the attributes country, country_code, income and region and joined with the country code to the GlobalFindexDatabase2025 dataset in a 1:* relationship. A table dimgender was created with values male and female, but was not linked with the GlobalFindexDatabase2025 dataset. The null values of Country, Income, and Region were filtered out. The following measures, namely, Account, BorrowAny, BorrowBank, BorrowFam, BorrowMobile, ComponentsPresent, CoveragePct, DebitCard, DFI, DFI Income Median, DFI Lift vs Income, DFI Rank(within income), DFI_raw_0to1, DigiAny, DigiAny Weighted, DigiMade, DigiRecv, DigitalPref01, EmergNotPossible, EmergPosssible, FinAccount, PopAdult, Rails, ResilienceCredit, SaveAny, SavedBank, SavedMobile, UtilityAll, UtilityCash, WageAcc, and WagePrivate were created by calculating their averages.

A total of five slicers, as seen in Figure 1, were created to filter across all the pages created, namely, Country, Gender, Income, Region, and Year. The null values of the attributes of the slicers were filtered out. These slicers would help filter across all the sub-categories, or a few of which are deemed necessary.

The KPIs(Key Performance Indicators), as seen in Figure 2, have been added so as not to have to go through all the other plots down the dashboard, but instead have the big picture of the whole dashboard. The KPI values include

- DFI - overall digital financial inclusion, this field is calculated using measures like general account, financial account, any digital payment, digital payment made and received, debit card use, digital preferences, payment rails index, resilience via savings or borrowing in different ratios equating to DFI_raw_0to1(decimal value) which then is multiplied by 100 to get to the percentage.
- Account% - shows the percentage of people with a bank or mobile account.
- DFI Income Median - Is the median DFI score of the country across the same income group.
- Any Digital payment% - It shows that people who have access to digital payment and are using it, as access does not translate to usage.
- Debit Card Ownership - If this is low, it implies that debit card ownership is low, as well as any form of digital payments are low, hence they are reliant on cash payment more and hence might have low digital knowledge.
- Any Saving% - whether they have any form of saving.

- Any Borrowing% - Reflects the credit reach, if it is high but the formal sources(banks or mobile money) are low, then the borrowing is most likely informal(family or friends) hence we could conclude that the credit limit could be expanded.
- Digital vs Cash% - is the ratio of cash to digital payments for essential bills.

A map was created for all the countries in the dataset as per Figure 3, and the legend was set to show the income of the people. From this map, details such as the measures account, any digital payments, any saved money, wage sent to the account, the Digital Financial Inclusion rate, and their digital preference rate can be inferred from the map.

The scatterplot in Figure 4 represents the percentage of digital payments made by people who received wages into their accounts. **X-axis** (WageAcc): % of adults (15+) who receive wages into an account, **Y-axis** (DigiAny): % of adults (15+) who made or received any digital payment, **bubble size**: Roughly represents population weight of that income group/country cluster, **bubble colour**: Income classification (High, Upper Middle, Lower Middle, Low).The comparison is made across all forms of household incomes. The tooltip includes information from measures like DFI and their digital preference rate.

From the table in Figure 5, it could be used to infer the DFI and their overall global rankings across the countries and all forms of household incomes.

The first bar plot, as shown in Figure 6, shows a comparison between the household incomes and the percentage of digital payments made in each subcategory. And the second bar graph as shown in Figure 7, shows the percentage of digital payments made across the regions. Both the bar plots have DFI in the tooltips.

All the visuals are interactive, and the slicers could be used to adjust the required metrics and get the preferred outputs.

Solutions:

After visualizing all the plots, KPIs and the table, we can infer the following for the global standards:

KPI:

- DFI (51.9) → The global digital financial inclusion level is moderate. If we put it on a 0–100 scale, we are just slightly above the halfway point for total and universal digital usage. But we are slightly above the median value which is 46.7 .
- Account (%) ≈ 60.5% → Approximately 60% of adults have bank accounts which means that access is good but not available everywhere.
- Digital vs Cash (%) ≈ 67.4% → In places where transactions take place, about 2/3 of them are in digital form. There is a change in preference, but the access and usage gaps are slowing down the global DFI from increasing faster.
- Debit Card Ownership (%) ≈ 47.6% → Less than half of the world's adult population owns a debit card. This goes hand in hand with many low-and lower-middle-income countries where mobile money is used instead of cards.

- Any Savings (%) ≈ 49.9% and Any Borrowing (%) ≈ 50.5% → Financial resilience is fragile: only about half saved or borrowed in the observed years (and a large chunk of borrowing is likely informal).
- Any Digital Payment (%) ≈ 53.4% → It is estimated that 50% of adults used digital payments during the last year.

Filled Map :

- High-income countries have been the most illuminated spots on the map with DFI values mostly between 70-90 with certain exceptions like Saudi Arabia, Uruguay, Chile, Panama, etc.
- The lowest DFI clusters (<35-50) are found in Sub-Saharan Africa and some areas of South Asia, Middle East and Northern Africa.
- The filled map draws attention to the inequality issue: the world DFI average (around 52) which is the case with some regions is very much polarized. Just a handful of high-income economies are responsible for the upward pull in the average, while most low-income countries are clustered below the 40 mark.

Scatter Plot :

- The trend is quite evident; with an increase in the amount of salaries that are paid via bank accounts, the usage of digital payment methods also goes up.
- Countries with high incomes (purple, top-right) → very high WageAcc (42.74%) and DigiAny (82.35%). It is the digital way of living in these regions.
- Countries in the upper-middle-income category (dark blue, mid-right) → WageAcc(22.88%), and DigiAny(50.87%). The market is huge for growth if wage digitisation makes a step forward.
- Countries in the lower-middle-income category (orange, lower-mid) → WageAcc(17.08%), DigiAny(39.14%). Limited access is the major reason for low digital uptake.
- Countries with low incomes (light blue, bottom-left) → very low WageAcc (13.25%) and DigiAny (33.12%). These are the most marginalized groups.
- The scatterplot is the evidence that wage-to-account digitisation is the most powerful global driver of financial inclusion.

Ranking Table :

- The average score of ~52.35 indicates a pretty good digital financial inclusion globally. This means only half the world's population is digitally included in financial transactions, although the use of digital accounts and transactions has increased.
- Australia (DFI 89.56, rank 1) ,Iceland (86.15,rank 2) and Switzerland(85.75,rank 3) are among the countries that rank very high in account ownership, have good wage-based and digital payment systems, and these are the features that characterize the digital financial inclusion(DFI) at its best. These High Income countries are the ones to follow, they show what is possible when modern infrastructures, regulations, and people's trust are combined.

- Upper Middle Income countries like Argentina (49.75, rank 72) and Belarus (61.35, rank 53) are close to the global average, which means they still have a long way to go regarding digital adoption.
- Lower Middle Income countries like Bangladesh (36.64,rank 108) and Bhutan (22.01, rank 149) are below the global average. In these countries, debit card growth has been a recent phenomenon (mainly due to mobile money), but still people do not trust, and poor digital infrastructure and cash economies are major deterrents to widespread use.
- Sudan (17.37, rank 152) and Chad (16.98, rank 153) share the lowest financial inclusion scores in the world. The lack of a reliable banking system, poor internet access, low literacy, and ongoing political and economic instability are all factors that keep these countries very far from the global average.
- On one hand, there is a digital divide between certain groups (urban, richer people) that use technology a lot and on the other hand, there are rural and poor people who hardly ever use it.

Bar Chart(DigiAny vs Income):

- High-income economies average ~82% of adults making/receiving a digital payment.
- Upper-middle drop to ~50%.
- Lower-middle fall further to ~39%.
- Low-income are lowest at ~33%.
- High Income have an ~2.5x higher rate in using digital payments compared to the lower income groups. The variance in the income groups explains the gap in the digital payments made.
- The "Access ≠ Use" gap between the incomes of different countries becomes rather negligible as they are on the same level of incomes since along with account ownership more and more active users start to appear in the area of usage (DigiAny) and thus leading to the conclusion of better rails higher trust and increased digitised wage/merchant flows.

Bar Chart(DigiAny vs Region):

- High-income group → 82.35%
- East Asia & Pacific (ex. high-income) → 50.63%
- Europe & Central Asia (ex. high-income) → 50.61%
- Sub-Saharan Africa → 42.24%
- Latin America & Caribbean → 41.3%
- South Asia → 32.68%
- Middle East & North Africa → 30.85%
- High-income countries are well and truly leading the charge: most adults now pay digitally, whereas lower-income economies continue to rely on cash more heavily.
- There are a few emerging countries that punch above their weight where governments and businesses made it easier for consumers to pay digitally.
- Finally, countries that allow wages and pensions to be deposited into a bank/mobile account saw significantly higher everyday digital use.

Conclusion:

The analysis shows that global financial inclusion issues are no longer just about access, but about empowerment and use. High-income economies show strong use of digital payments, savings and wage digitization, while low- and middle-income countries are still lagging, due to reliance on cash, lack of strong digital payments infrastructure, and socio-economics.

Using the methods we examined in the unit (data cleaning, creating compound measures such as DFI, visual storytelling in Power BI) the study shows:

From the data: income and region are strong predictors of adoption, but structural enablers like wage digitization can help accelerate the use of some digital services even in low-income countries.

From visual designing: Building complex compound indices (like DFI) and using benchmarks (rank within income) give fairer comparisons across countries and regions.

From visual storytelling: dashboards show not only where gaps exist but also where targeted policy actions like wage digitization or interoperability can have impact.

In practice, the findings can be used by states and financial institutions not just to move beyond access but towards use, inclusion of rural populations, and financial empowerment in the longer-term.

Discussion :

- Initially, there was some uncertainty over the best approach to manage nulls, whether to "impute" the nulls or just drop them. However, the issue slowly resolved when I built dimension tables (country, year, income, gender) and made sure to set relationships in Power BI clearly at a later stage, and there were no longer more nulls as they were filtered out while building the visuals.
- The relationships between tables (fact vs dimension), were not actively connected, which resulted in errors occurring on measures. The readjustment of the dimensions into a nice star schema with the correct entity relationships made all the difference in establishing clear connections (country - year - allmeasures).
- The main ambiguity while developing the project’s problem statement was whether digital financial inclusion is mostly an 'access' problem or a 'usage' problem. Through the data analysis process, it became very evident that although there has been a positive trend towards increased account access globally, the gap that really matters is active use, especially in low and middle-income countries. Ultimately, through building BI dashboards and DFI scorecards, I was able to answer the question with evidence (with and without measures) and demonstrate which parts of the income - region - wage digitisation, are driving usage differences around the world.

Appendix :

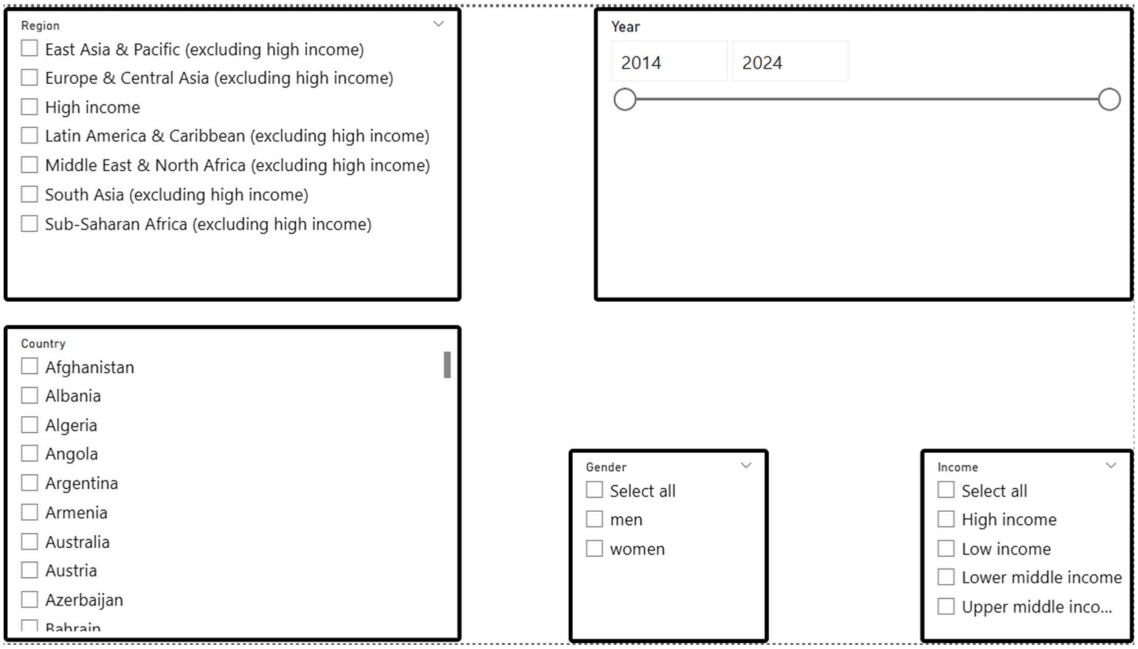

Fig 1 : Slicers

Fig 2 : Key Performance Indicators

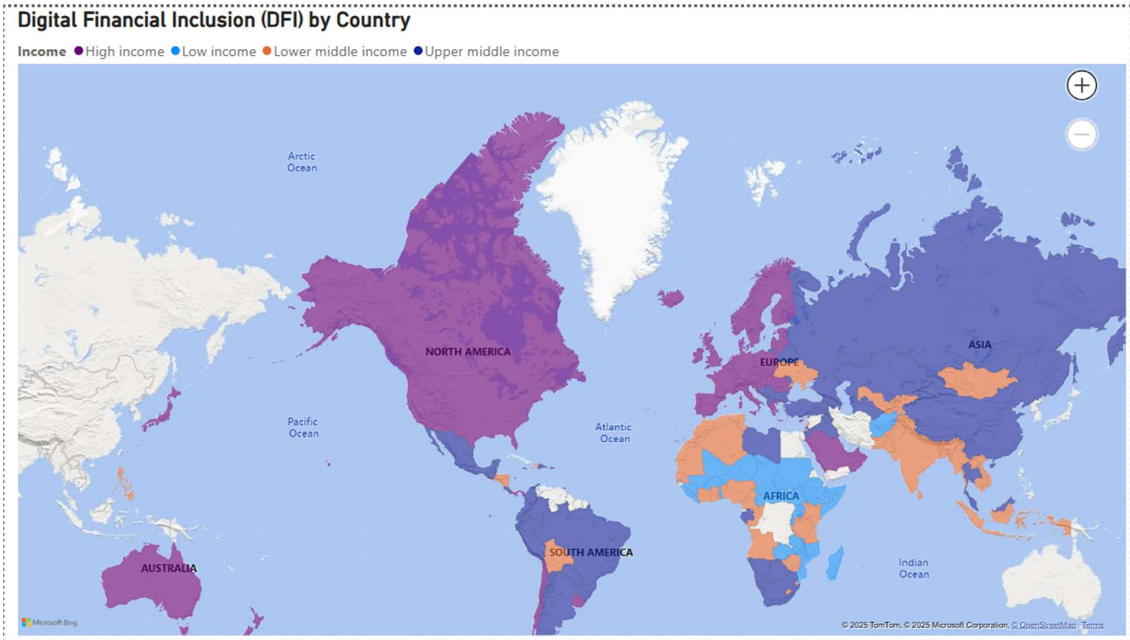

Fig 3 : Filled Map(DFI vs Country)

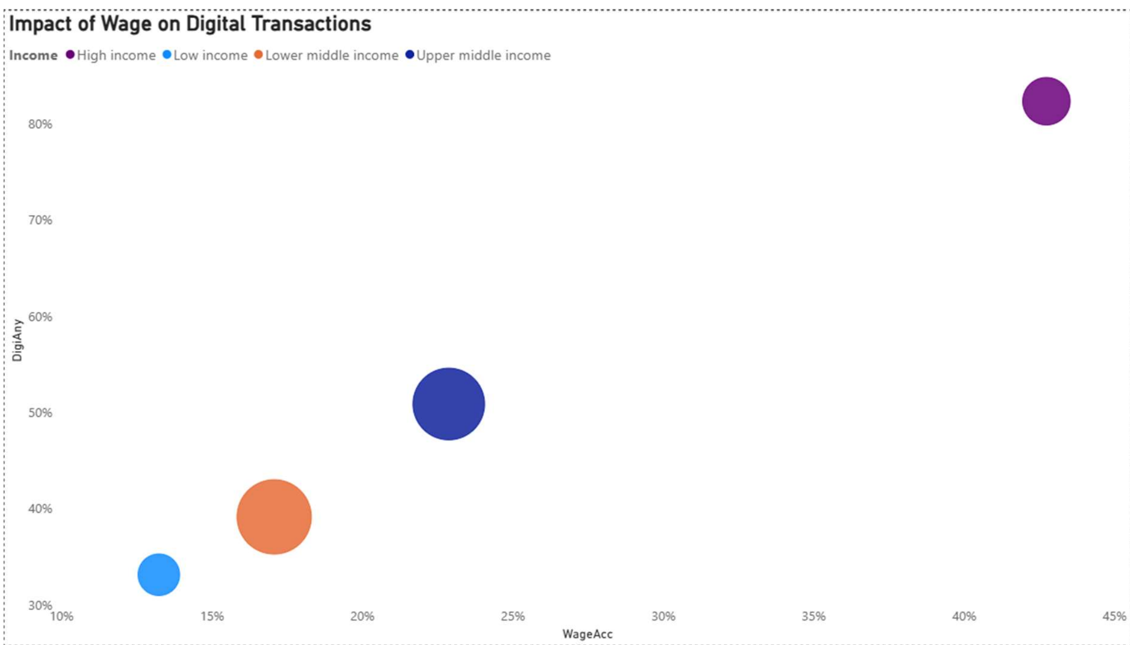

Fig 4 : Scatter Plot(Digital Use)

DFI Scores and Rankings by Country and Income Group

Country	Income	DFI	DFI Rank (within income)
Afghanistan	Low income		154
Albania	Upper middle income	30.18	131
Algeria	Lower middle income	31.90	123
Angola	Lower middle income	31.87	124
Argentina	Upper middle income	49.75	72
Armenia	Upper middle income	41.18	94
Australia	High income	89.56	1
Austria	High income	77.83	24
Azerbaijan	Upper middle income	34.05	119
Bahrain	High income	67.34	42
Bangladesh	Lower middle income	36.64	108
Belarus	Upper middle income	61.35	53
Belgium	High income	78.57	23
Belize	Upper middle income	43.29	87
Benin	Lower middle income	39.24	99
Bhutan	Lower middle income	22.01	149
Total		**52.35**	**69**

Fig 5 : Table(Rankings)

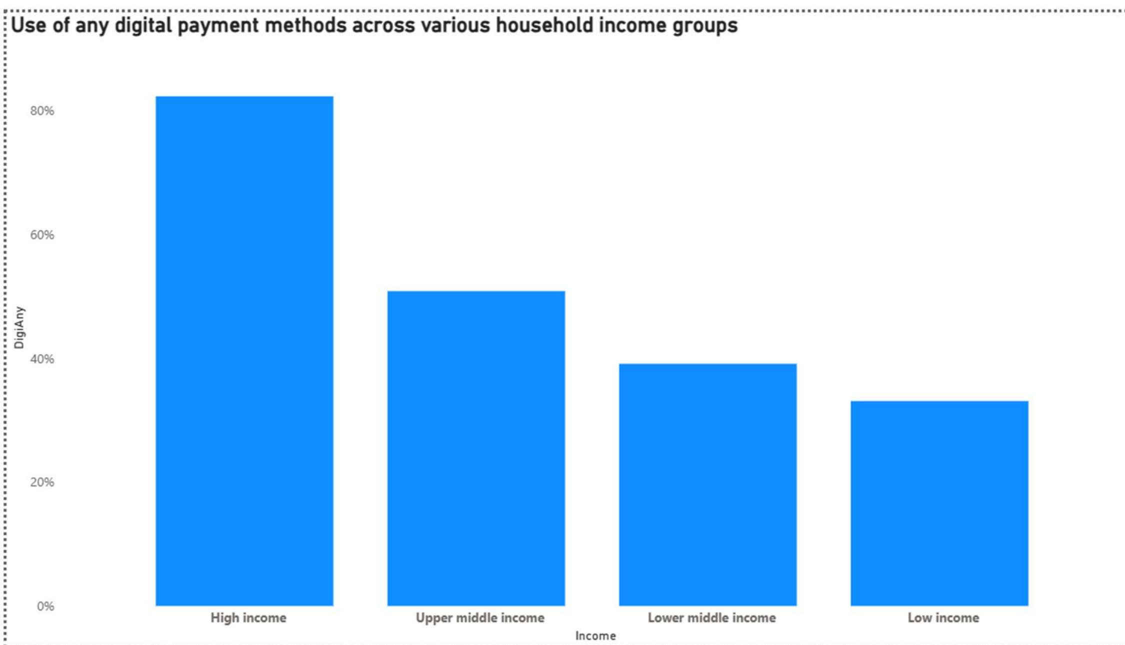

Fig 6 : Bar Chart(DigiAny vs Income)

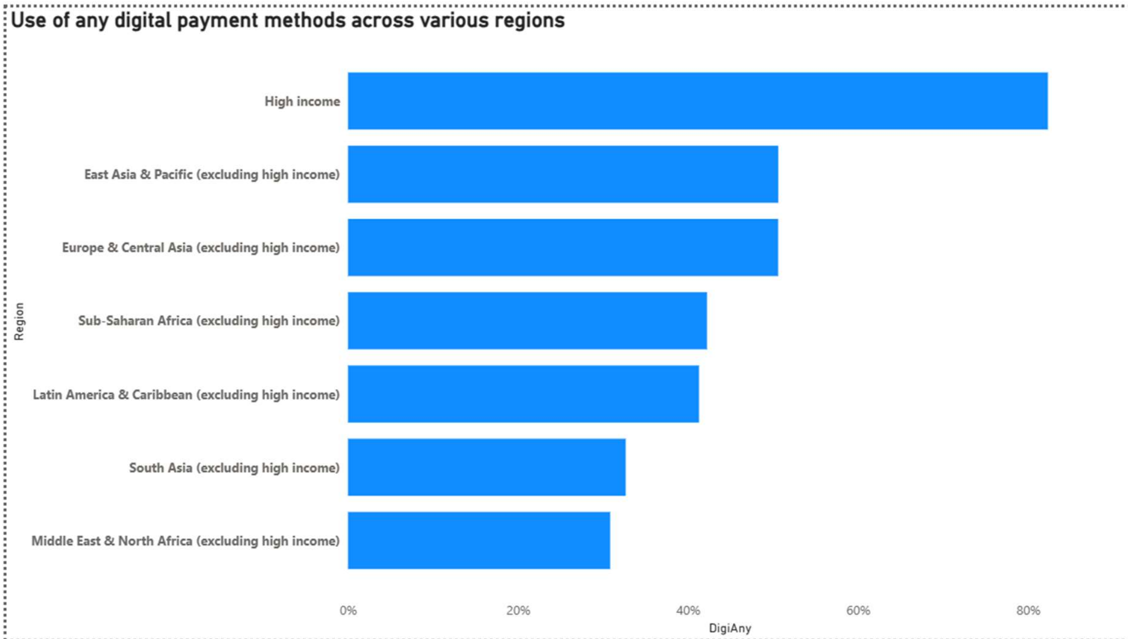

Fig 7 : Bar Chart(DigiAny vs Region)

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